

Package ‘uSDT’

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Type Package

Title Hierarchical Signal Detection Theory Models for Unconscious Processing

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Description Fits hierarchical signal detection theory (SDT) models to paired direct and indirect measures, the design used to test for unconscious processing. Continuous indirect measures (typically response times) are dichotomized with the within-subject median split of Meyen et al. (2022) <[doi:10.1037/xge0001065](https://doi.org/10.1037/xge0001065)> so that both tasks are placed on a common sensitivity scale. The package estimates a binomial probit mixed model in which the two sensitivities are correlated random effects, and tests the three hypotheses of interest: the group-level difference between sensitivities, their latent correlation, and the latent regression of the indirect on the direct measure, whose intercept is the test for unconscious processing. Frequentist estimation uses 'lme4'.

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Author Ricardo Rey-Sáez [aut, cre] (ORCID:
<https://orcid.org/0000-0001-6739-2035>),
Francisco Garre-Frutos [aut] (ORCID:
<https://orcid.org/0000-0001-9810-186X>),
Alicia Franco-Martínez [aut] (ORCID:
<https://orcid.org/0000-0002-9710-1240>),
Miguel Vadillo [aut] (ORCID: <https://orcid.org/0000-0001-8421-816X>)
Maintainer Ricardo Rey-Sáez <ricardoreysaez95@gmail.com>
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hsdt	<i>Fit a hierarchical signal detection theory model</i>
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Description

Fits a bivariate hierarchical SDT model using `lme4::glmer()` and evaluates the core unconscious processing hypotheses. The model estimates task-specific sensitivities (d') and response criteria (c) as fixed effects, while estimating their variation and correlation across participants via random effects.

Usage

```
hsdt(  
  data,  
  estimation = c("frequentist"),  
  fix_criteria = c("auto", "none"),  
  level = 0.95,  
  optimizer = "bobyqa",  
  ...  
)
```

```
## S3 method for class 'hsdt'
summary(object, ...)

## S3 method for class 'hsdt'
print(x, ...)
```

Arguments

<code>data</code>	A <code>usdt_data</code> object from <code>usdt_data_tasks()</code> or <code>usdt_data_long()</code> .
<code>estimation</code>	Estimation framework. Currently only "frequentist" (maximum likelihood via Laplace approximation) is supported.
<code>fix_criteria</code>	How to handle response criteria. "auto" fixes to zero any criterion that is zero by design (such as a task split at the median under deviation coding). "none" estimates all criteria.
<code>level</code>	Confidence level for Wald intervals (default is 0.95).
<code>optimizer</code>	Primary optimizer passed to <code>lme4::glmerControl()</code> . Alternative optimizers are automatically evaluated if the default fails to converge or produces a singular fit.
<code>...</code>	Additional arguments passed to <code>lme4::glmer()</code> . Model formula, family, and data inputs remain managed by the package.
<code>object</code>	An <code>hsdt</code> object.
<code>x</code>	An <code>hsdt</code> object.

Details

The model fits trial counts with a binomial probit link, directly mapping coefficients to standard Signal Detection Theory parameters. Fixed effects capture population sensitivities and criteria, while random effects estimate participant variation and the latent correlation between direct and indirect sensitivity.

Hypotheses evaluated by default:

- **H1:** Mean sensitivity difference between tasks.
- **H2:** Latent correlation of sensitivities across participants.
- **H3:** Latent regression of indirect on direct sensitivity. Its intercept reflects expected indirect performance when direct awareness is zero ($d'_{\text{Direct}} = 0$).

When sample sizes or trial counts are low, variance components can reach singular boundaries. In these cases, the function issues a warning, and parametric bootstrap intervals can be calculated using `usdt_boot()`.

Value

An object of class `hsdt` containing:

- `$fit`: The underlying `glmerMod` object from `lme4`.
- `$tests`: Summary table for hypotheses H1, H2, and H3.

- `$pars`: Model parameter estimates on the SDT scale.
- `$design`: Summary of the model specification and formula.
- `$diagnostics`: Convergence flags and singular fit indicators.

See Also

[usdt_data_tasks\(\)](#), [usdt_tests\(\)](#), [usdt_boot\(\)](#), [plot.hsdt\(\)](#)

Examples

```
# Contextual cuing data from Vadillo et al. (2025)
d <- usdt_data_tasks(
  direct    = vadillo_awareness,
  indirect  = vadillo_cuing,
  subject_col = "subj",
  condition_col = "condition",
  condition_levels = c(signal = "old", noise = "new"),
  response_col = list(direct = "judged.old", indirect = "rt"),
  response_levels = list(direct = c(signal = 1, noise = 0),
                        indirect = c(signal = "faster", noise = "slower")),
  dichotomize = list(direct = FALSE, indirect = TRUE)
)

m <- hsdt(d)

# Full summary table with SDT parameters and hypothesis tests
summary(m)

# Inspect the model formula (indirect criterion omitted by default)
m$design$formula
```

meyen_split

Dichotomize response times into binary choices

Description

Splits response times (or other continuous measures) at each subject's overall median, following the preprocessing approach of Meyen et al. (2022). The median is calculated across all trials for each participant without distinguishing between stimulus conditions or other covariates. This produces a binary outcome that allows response times to be mapped onto a Signal Detection Theory sensitivity metric (d').

Usage

```
meyen_split(x, by, signal = c("faster", "slower"), ties = c("noise", "random"))
```

Arguments

<code>x</code>	Numeric vector of continuous values, typically response times.
<code>by</code>	Vector identifying the subject for each observation in <code>x</code> . Medians are computed independently for each participant.
<code>signal</code>	Character string specifying which side of the median will be treated as the "signal" response under an SDT framework. Use "faster" when the target condition speeds up responses (e.g., facilitatory priming, spatial cueing) or "slower" when it slows responses down (e.g., interference, Stroop-like effects).
<code>ties</code>	How to handle trials that match the subject's median exactly. "noise" assigns them to the noise category (0). "random" breaks ties at random, keeping cell proportions as balanced as possible.

Details

With an odd number of trials (n), a dataset cannot be split into two equal halves because the median falls exactly on an observed trial. Setting `ties = "noise"` assigns this middle trial to noise, producing a signal proportion of $(n - 1)/(2 \cdot n)$ and slightly shifting the response criterion. In practice, this difference ($1/(2 \cdot n)$) is negligible, but setting `ties = "random"` resolves ties probabilistically to avoid any systematic directional bias.

Value

An integer vector of 0s (noise response) and 1s (signal response) matching the length of `x`. Missing values (NA) are preserved.

References

Meyen, S., Zerweck, I. A., Amado, C., von Luxburg, U., & Franz, V. H. (2022). Advancing research on unconscious priming: When can scientists claim an indirect task advantage? *Journal of Experimental Psychology: General*, 151(1), 65–81. doi:10.1037/xge0001065

Examples

```
rt <- c(320, 410, 295, 500, 380, 450)
subj <- rep(c("s1", "s2"), each = 3)
meyen_split(rt, by = subj)
```

Description

Generates diagnostic visualizations for a fitted hierarchical SDT model, including observed versus latent regression (H3), shrinkage patterns, participant-level caterpillar intervals, and model-implied ROC curves.

Usage

```
## S3 method for class 'hsdt'
plot(
  x,
  type = c("regression", "shrinkage", "caterpillar", "roc"),
  subject_id = NULL,
  band = TRUE,
  population_reference = TRUE,
  observed_se = NULL,
  ...
)
```

Arguments

x	An hsdt object fitted by <code>hsdt()</code> .
type	Character string indicating the plot type: • "regression": Compares the observed OLS regression with the latent regression line (H3). • "shrinkage": Connects observed d' to model-implied d' for each participant across bivariate contours. • "caterpillar": Compares observed d' and model-implied d' with confidence intervals for each participant alongside zero. • "roc": Shows model-implied ROC curves with estimated criteria.
subject_id	Identifier for a specific participant when <code>type = "roc"</code> . If omitted, displays population-level curves.
band	Logical. Show confidence bands around regression lines or ROC curves (default is TRUE).
population_reference	Logical. When plotting an individual ROC curve, add population-average curves as dashed reference lines.
observed_se	Variance formulation for empirical intervals in "caterpillar" plots: "gg" (Gourevitch & Galanter, 1967, default) or "miller" (Miller, 1996).
...	Additional arguments passed to underlying plotting methods.

Value

A ggplot object. Its underlying data frame is stored in `$data`.

Regression plot (type = "regression")

Compares observed and latent associations across two panels sharing axes. The left panel shows observed d' values and an ordinary least-squares line. When direct task reliability is low, trial-level sampling noise attenuates this observed slope toward zero. The right panel plots the latent regression line (d'_I on d'_D) from H3, correcting for measurement error. The value of each line at $d'_D = 0$ marks the intercept testing for unconscious processing, and both panels display it the same way: an open circle at the point estimate with a vertical line spanning its confidence interval. The observed marker

is the least-squares intercept and the latent marker is its measurement-error-corrected counterpart, so the two panels place the same hypothesis side by side. Confidence bands are computed via the delta method or bootstrap replicates when `usdt_boot()` is present.

Shrinkage plot (type = "shrinkage")

Connects each participant's observed d' (from `sdt_moments()` with Hautus correction) to their model-implied d' . The lower the reliability of the measures, the higher the shrinkage of observed estimates toward the group-level mean.

Caterpillar plot (type = "caterpillar")

Plots observed d' and model-implied d' with confidence intervals for every participant. Empirical intervals use normal approximations based on `observed_se`. Model-implied intervals incorporate uncertainty from population means, variance components, and participant random effects.

ROC plot (type = "roc")

Displays model-implied ROC curves for an average participant or a specific individual, with points marking the estimated response criteria.

See Also

`hsdt()`, `sdt_moments()`, `usdt_boot()`

Examples

```
# Contextual cuing data from Vadillo et al. (2025)
d <- usdt_data_tasks(
  direct   = vadillo_awareness,
  indirect = vadillo_cuing,
  subject_col = "subj",
  condition_col = "condition",
  condition_levels = c(signal = "old", noise = "new"),
  response_col = list(direct = "judged.old", indirect = "rt"),
  response_levels = list(direct = c(signal = 1, noise = 0),
                        indirect = c(signal = "faster", noise = "slower")),
  dichotomize = list(direct = FALSE, indirect = TRUE)
)

fit <- hsdt(d)

# 1. Observed vs. latent regression (H3)
plot(fit, type = "regression")

# 2. Bivariate shrinkage toward the group mean
plot(fit, type = "shrinkage")

# 3. Participant-level intervals (observed vs. model-implied)
plot(fit, type = "caterpillar")

# 4. Model-implied ROC curves
```

```
plot(fit, type = "roc")
plot(fit, type = "roc", subject_id = "2001", population_reference = TRUE)
```

sdt_moments

Signal detection measures for each subject

Description

Computes empirical hit rates, false-alarm rates, sensitivity (d'), and response criteria for each participant without fitting a model. It can also calculate sampling variances, standard errors, and expected values for d' .

Usage

```
sdt_moments(
  data,
  subject_col = NULL,
  condition_col = NULL,
  condition_levels = NULL,
  response_col = NULL,
  response_levels = NULL,
  coding = c("deviation", "treatment"),
  correction = c("hautus", "none"),
  variances = FALSE
)
```

Arguments

data	A usdt_data object or a standard trial-level data frame. When given a usdt_data object, the function processes both tasks and includes a task column in the output.
subject_col, condition_col, response_col	Column names for subject, condition, and response variables. Only required when data is a plain data frame.
condition_levels, response_levels	Named vectors mapping condition and response labels, like c(signal = "old", noise = "new"). Required for a plain data frame. A usdt_data object supplies its own roles and needs neither.
coding	Criterion definition to report: "deviation" measures the criterion from the mid-point between the signal and noise distributions, whereas "treatment" measures it from the noise distribution. A usdt_data object supplies its own coding.
correction	Handling of extreme rates (0 or 1) that make d' infinite. "hautus" adds 0.5 to all four cell counts for affected participants. "none" leaves infinite values in place.
variances	Logical. If TRUE, computes the sampling variance of d' from Gourevitch and Galanter (1967) and Miller (1996), each with its own standard error, as well as the expected value of d' under Miller's distribution.

Details

Edge corrections apply only to participants with extreme rates (0 or 1) rather than the whole sample, leaving well-defined rates unchanged.

When requested, the sampling variance of d' is estimated using the asymptotic approximation of Gourevitch and Galanter (1967) and the binomial-distribution approach of Miller (1996). See Suero et al. (2017) for a comparison between the two approaches.

Value

A data frame with one row per subject (or per subject and task for `usdt_data` inputs) containing:

- `hit`, `miss`, `fa`, `cr`: Raw response counts.
- `hr`, `far`: Observed hit and false-alarm rates.
- `zhr`, `zfar`: Probit-transformed rates.
- `dprime`, `criterion`: Descriptive SDT estimates.
- `corrected`: Logical flag indicating whether the participant received an edge correction.
- `var_gg`, `se_gg`: Asymptotic variance and standard error from Gourevitch and Galanter (1967), present when `variances = TRUE`.
- `var_miller`, `se_miller`, `expected_dprime`: Moments from Miller (1996), present when `variances = TRUE`.

References

Gourevitch, V., & Galanter, E. (1967). A significance test for one parameter isosensitivity functions. *Psychometrika*, 32(1), 25–33. doi:[10.1007/BF02289402](https://doi.org/10.1007/BF02289402)

Hautus, M. J. (1995). Corrections for extreme proportions and their biasing effects on estimated values of d' . *Behavior Research Methods, Instruments, & Computers*, 27(1), 46–51. doi:[10.3758/BF03203619](https://doi.org/10.3758/BF03203619)

Miller, J. (1996). The sampling distribution of d' . *Perception & Psychophysics*, 58(1), 65–72. doi:[10.3758/BF03205476](https://doi.org/10.3758/BF03205476)

Suero, M., Privado, J., & Botella, J. (2017). Methods to estimate the variance of some indices of the signal detection theory: A simulation study. *Psicologica*, 38(1), 77–109.

See Also

[hsdt\(\)](#)

Examples

```
# 1. From a prepared usdt_data object (both tasks at once)
d <- usdt_data_tasks(
  direct    = vadillo_awareness,
  indirect  = vadillo_cuing,
  subject_col = "subj",
  condition_col = "condition",
  condition_levels = c(signal = "old", noise = "new"),
  response_col = list(direct = "judged.old", indirect = "rt"),
```

```

response_levels = list(direct = c(signal = 1, noise = 0),
                        indirect = c(signal = "faster", noise = "slower")),
dichotomize     = list(direct = FALSE, indirect = TRUE)
)

head(sdt_moments(d))

# 2. From raw trials with sampling variances and standard errors
head(sdt_moments(vadillo_awareness,
                 subject_col = "subj",
                 condition_col = "condition",
                 condition_levels = c(signal = "old", noise = "new"),
                 response_col = "judged.old",
                 response_levels = c(signal = 1, noise = 0),
                 variances = TRUE))

```

usdt_boot

*Parametric bootstrap intervals for hierarchical SDT models***Description**

Simulates new datasets from a fitted model using `lme4::bootMer()`, refits the model to each replicate, and computes bootstrap confidence intervals for the three core hypotheses (H1, H2, H3). This is especially useful when asymptotic Wald intervals are unreliable due to singular or boundary fits.

Usage

```

usdt_boot(
  object,
  nsim = 1000,
  ncores = 1L,
  max_attempts = 2 * nsim,
  seed = NULL,
  progress = interactive(),
  level = 0.95,
  type = c("perc", "norm", "basic")
)

```

Arguments

<code>object</code>	An <code>hsdt</code> object fitted by <code>hsdt()</code> .
<code>nsim</code>	Target number of successful replicates (at least 500).
<code>ncores</code>	Number of CPU cores for parallel processing. Values above 1 create a temporary cluster that works across Windows, macOS, and Linux, and automatically stops when finished.
<code>max_attempts</code>	Maximum number of refits to attempt. Defaults to $2 * nsim$.

seed	Random seed for reproducibility. With the default NULL, no seed is set and the bootstrap continues the current random number stream.
progress	Logical. Display a progress bar during fitting (defaults to TRUE in interactive sessions).
level	Confidence level for intervals (default is 0.95).
type	Type of bootstrap interval: "perc" (percentile), "norm" (normal approximation with bias correction), or "basic" (empirical basic). For correlations, "norm" and "basic" operate on the Fisher- z scale; if the sample correlation lies on the boundary (-1 or 1), these types return NA, whereas "perc" remains available.

Details

Replicates are dropped only if the model fails to fit or does not converge. Singular fits and boundary estimates are intentionally retained because discarding them artificially narrows intervals in constrained settings.

When an attempted run finishes with fewer than 500 usable replicates, the original Wald intervals are preserved, a warning is issued, and raw attempt diagnostics are stored in `$boot`.

Point estimates remain identical to the original model fit. Two-sided bootstrap p -values compare the observed test statistic against the centered bootstrap distribution using standard finite-sample adjustment $((k + 1)/(B + 1))$, ensuring p -values never equal zero.

Value

An updated `hsdt` object where interval columns in `$tests` are replaced by bootstrap estimates. A new `$boot` element contains:

- `$t`: Matrix of replicates for the three hypotheses.
- `$variance`: Replicates of sensitivity variances.
- `$population`: Replicates of average criteria and sensitivities.
- `$subjects`: Replicates of individual-level parameters.
- Run diagnostics and convergence counts (usable, attempted, retained, failures).

See Also

`hsdt()`, `usdt_tests()`

Examples

```
# Contextual cuing data from Vadillo et al. (2025)
d <- usdt_data_tasks(
  direct = vadillo_awareness,
  indirect = vadillo_cuing,
  subject_col = "subj",
  condition_col = "condition",
  condition_levels = c(signal = "old", noise = "new"),
  response_col = list(direct = "judged.old", indirect = "rt"),
  response_levels = list(direct = c(signal = 1, noise = 0),
                        indirect = c(signal = "faster", noise = "slower")),
```

```

    dichotomize      = list(direct = FALSE, indirect = TRUE)
  )

  m <- hsdt(d)

  # Run parametric bootstrap with 500 replicates. Refitting this model 500
  # times takes several minutes
  b <- usdt_boot(m, nsim = 500, seed = 1)

  # Inspect updated summary with bootstrap intervals and p-values
  summary(b)

  # Check fit diagnostics across bootstrap replicates
  b$boot[c("usable", "attempted", "retained", "failures")]

```

usdt_data

Prepare data for hierarchical SDT models

Description

Formats direct and indirect task data into a standardized structure for `hsdt()`. Use `usdt_data_tasks()` when tasks are stored in separate data frames, or `usdt_data_long()` when both tasks are kept in a single data frame with a column that identifies each task.

Usage

```

usdt_data_tasks(
  direct,
  indirect,
  subject_col,
  condition_col = NULL,
  condition_levels = NULL,
  response_col = NULL,
  response_levels = NULL,
  successes_col = NULL,
  trials_col = NULL,
  successes_type = c("auto", "counts", "proportions"),
  sdt_cols = NULL,
  dichotomize = c("none", "indirect", "direct", "both"),
  ties = c("noise", "random"),
  coding = c("deviation", "treatment"),
  labels = c(direct = "Direct", indirect = "Indirect")
)

usdt_data_long(

```

```

    data,
    task_col,
    task_levels,
    subject_col,
    condition_col = NULL,
    condition_levels = NULL,
    response_col = NULL,
    response_levels = NULL,
    successes_col = NULL,
    trials_col = NULL,
    successes_type = c("auto", "counts", "proportions"),
    sdt_cols = NULL,
    dichotomize = c("none", "indirect", "direct", "both"),
    ties = c("noise", "random"),
    coding = c("deviation", "treatment"),
    labels = NULL
)

## S3 method for class 'usdt_data'
print(x, ...)

```

Arguments

<code>direct, indirect</code>	Data frames for each task (used in <code>usdt_data_tasks()</code>).
<code>subject_col</code>	Name of the column that identifies participants.
<code>condition_col</code>	Name of the column for signal and noise conditions. Not needed when using <code>sdt_cols</code> .
<code>condition_levels</code>	Named vector mapping condition labels, like <code>c(signal = "old", noise = "new")</code> . Required: which label is the signal and the noise.
<code>response_col</code>	Name of the column with responses. Can be binary choices or continuous values (like response times) to split at the median.
<code>response_levels</code>	Named vector mapping responses, like <code>c(signal = 1, noise = 0)</code> . For a task named in <code>dichotomize</code> it names the side of the median instead, as <code>c(signal = "faster", noise = "slower")</code> or the reverse. Required in both cases.
<code>successes_col, trials_col</code>	Names of columns with pre-calculated counts or proportions of signal responses and total trials. Use these instead of <code>response_col</code> .
<code>successes_type</code>	Format of <code>successes_col</code> : "counts", "proportions", or "auto".
<code>sdt_cols</code>	Named vector for SDT table columns, like <code>c(hit = "H", miss = "M", fa = "FA", cr = "CR")</code> .
<code>dichotomize</code>	Which tasks to split at the median using meyen_split() . Use "none", "direct", "indirect", "both", or a list like <code>list(direct = FALSE, indirect = TRUE)</code> .
<code>ties</code>	How to handle trials that fall exactly on the median. See meyen_split() .

coding	How condition is coded in the model: "deviation" (-0.5, 0.5) or "treatment" (0, 1). See Details.
labels	Optional names for the tasks in printed output.
data	A single data frame with both tasks (used in <code>usdt_data_long()</code>).
task_col	Name of the column that identifies the task in data.
task_levels	Named vector mapping task labels, like <code>c(direct = "D", indirect = "I")</code> .
x	A <code>usdt_data</code> object.
...	Ignored.

Details

The functions count responses for each subject and condition, check that the same subjects appear in both tasks, and print a summary table so you can verify the column settings before fitting the model.

Value

An object of class `usdt_data`. The `$agg` table contains the counts used by `hsdt()`, and `$meta` contains setup details and summaries.

Settings per task

Arguments for columns and levels take either a single value (used for both tasks) or a list with separate settings for each task:

```
condition_col    = list(direct = "cond", indirect = "cue")
condition_levels = list(direct  = c(signal = "old", noise = "new"),
                        indirect = c(signal = "congruent", noise = "incongruent"))
dichotomize      = list(direct = FALSE, indirect = TRUE)
```

This works for all column and level arguments, so you can combine trial-level data in one task with summary tables in the other.

Condition coding

Under "deviation" coding (-0.5 vs. +0.5), the intercept is $-c$, the criterion measured from the point between the two distributions. Under "treatment" coding (0 vs. 1), the intercept is $z(\text{FAR})$, the criterion measured from the noise distribution.

When a task is split at the median, deviation coding sets the group criterion to zero in balanced designs, so the model does not need to estimate it.

See Also

`hsdt()`, `me yen_split()`, `sdt_moments()`

Examples

```
# 1. Tasks in separate data frames
# Direct task: binary choices (old/new)
# Indirect task: response times (split at the median)
d_separate <- usdt_data_tasks(
  direct    = vadillo_awareness,
  indirect  = vadillo_cuing,
  subject_col = "subj",
  condition_col = "condition",
  condition_levels = c(signal = "old", noise = "new"),
  response_col = list(direct = "judged.old", indirect = "rt"),
  response_levels = list(direct = c(signal = 1, noise = 0),
                        indirect = c(signal = "faster", noise = "slower")),
  dichotomize = list(direct = FALSE, indirect = TRUE)
)
d_separate

# 2. Tasks combined in a single long data frame
long <- rbind(
  data.frame(task      = "D",
             subj      = vadillo_awareness$subj,
             condition = vadillo_awareness$condition,
             response   = vadillo_awareness$judged.old),
  data.frame(task      = "I",
             subj      = vadillo_cuing$subj,
             condition = vadillo_cuing$condition,
             response   = vadillo_cuing$rt)
)

d_long <- usdt_data_long(
  long,
  task_col      = "task",
  task_levels   = c(direct = "D", indirect = "I"),
  subject_col   = "subj",
  condition_col = "condition",
  condition_levels = c(signal = "old", noise = "new"),
  response_col  = "response",
  response_levels = list(direct = c(signal = 1, noise = 0),
                        indirect = c(signal = "faster", noise = "slower")),
  dichotomize   = list(direct = FALSE, indirect = TRUE)
)
d_long
```

usdt_hypotheses

Test the three core hypotheses of a hierarchical SDT model

Description

Evaluates the difference between average task sensitivities (H1), their correlation across subjects (H2), and the regression of indirect sensitivity on direct sensitivity (H3). `usdt_tests()` computes

all three together, while individual functions compute them separately.

Usage

```
usdt_tests(fit, direct = "d_D", indirect = "d_I", level = 0.95)

sensitivity_diff(fit, direct = "d_D", indirect = "d_I", level = 0.95)

latent_cor(fit, direct = "d_D", indirect = "d_I", level = 0.95)

latent_regression(fit, direct = "d_D", indirect = "d_I", level = 0.95)
```

Arguments

<code>fit</code>	A fitted model: an <code>hsdt</code> object or a <code>glmerMod</code> from <code>lme4::glmer()</code> .
<code>direct, indirect</code>	Character strings naming the sensitivity terms in the model. Defaults match the internal naming of <code>hsdt()</code> . For custom models, both terms must be fixed effects and share a common random-effects grouping by subject.
<code>level</code>	Confidence level for intervals (default is 0.95).

Details

These tests run automatically inside `hsdt()` and appear in its summary. Calling them directly is especially useful when fitting custom models with `lme4::glmer()`, allowing you to test these hypotheses while controlling for additional covariates (e.g., set size, experimental groups).

Value

A data frame with columns `term`, `estimate`, `se`, `statistic`, `p.value`, `conf.low`, `conf.high`, and `ci_method`. Columns `status` and `reason` flag unsupported estimates (e.g., singular fits). `usdt_tests()` includes an extra hypothesis column (H1, H2, H3).

The three hypotheses

- **H1 (Mean difference):** Tests whether average sensitivity differs between the direct and indirect tasks.
- **H2 (Correlation):** Tests the correlation between task sensitivities across participants using a Fisher-*z* transformed interval.
- **H3 (Latent regression):** Regresses indirect sensitivity onto direct sensitivity. The intercept represents expected indirect performance when direct sensitivity is zero ($d'_{\text{Direct}} = 0$), testing for unconscious processing.

Because both the correlation (H2) and regression slope (H3) are zero if and only if the covariance between sensitivities is zero, they evaluate the same association and share identical test statistics.

Custom models with covariates

To adjust tests for additional factors, specify the model directly using `lme4::glmer()`. As long as the two sensitivity terms are included as fixed effects and correlated across subjects via random slopes, `usdt_tests()` will compute the latent tests conditional on those covariates.

See Also

`hsdt()`, `usdt_boot()`

Examples

```
# 1. Standard model via hsdt()
d <- usdt_data_tasks(
  direct    = vadillo_awareness,
  indirect  = vadillo_cuing,
  subject_col = "subj",
  condition_col = "condition",
  condition_levels = c(signal = "old", noise = "new"),
  response_col = list(direct = "judged.old", indirect = "rt"),
  response_levels = list(direct = c(signal = 1, noise = 0),
                        indirect = c(signal = "faster", noise = "slower")),
  dichotomize = list(direct = FALSE, indirect = TRUE)
)

m <- hsdt(d)

# All three tests at once
usdt_tests(m)

# Or one test at a time
sensitivity_diff(m) # H1
latent_cor(m)       # H2
latent_regression(m) # H3

# 2. Custom model with covariates via glmer()
# Controlling for display set size across both tasks
trials <- rbind(
  data.frame(vadillo_awareness[c("subj", "condition", "set.size")],
    task = "D", resp = vadillo_awareness$judged.old),
  data.frame(vadillo_cuing[c("subj", "condition", "set.size")],
    task = "I", resp = meyen_split(vadillo_cuing$rt,
                                   by = vadillo_cuing$subj))
)

# Deviation contrasts (-0.5 vs 0.5); `direct` flags the direct task
trials <- within(trials, {
  cond <- ifelse(condition == "old", 0.5, -0.5)
  size <- ifelse(set.size == "set size 16", 0.5, -0.5)
  direct <- as.integer(task == "D")
})
```

```
# Standard glmer formula: indirect criterion is omitted (fixed at 0
# by the median split). Random effects estimate the direct criterion
# and correlated task sensitivities across subjects.
fit <- lme4::glmer(
  resp ~ 0 + direct + task:size + task:cond +
    (0 + direct | subj) + (0 + task:cond | subj),
  data = trials, family = binomial("probit"),
  control = lme4::glmerControl(optimizer = "bobyqa")
)

# Check the names lme4 assigned to the sensitivity terms
names(lme4::fixef(fit))

# Evaluate hypotheses conditional on set size
usdt_tests(fit, direct = "taskD:cond", indirect = "taskI:cond")
```

usdt_reliability	<i>Reliability of direct and indirect task measures</i>
------------------	---

Description

Estimates the reliability of sensitivity (d') by separating true variance across participants from sampling noise caused by finite trial counts. Values close to 1 indicate that the measure reliably separates participants, whereas values near 0 indicate that observed differences are mostly measurement error.

Usage

```
usdt_reliability(object)

## S3 method for class 'usdt_reliability'
summary(object, ...)

## S3 method for class 'usdt_reliability'
print(x, ...)
```

Arguments

object	An hsdtd object from <code>hsdt()</code> , which may also contain bootstrap results from <code>usdt_boot()</code> .
...	Ignored.
x	A usdt_reliability object.

Details

For participant i in task j , reliability is defined as:

$$\frac{\tau_j^2}{\tau_j^2 + v_{ij}}$$

where τ_j^2 is the true variance in sensitivity across participants from the model's random effects, and v_{ij} is the squared standard error of d' for that participant. This error variance reflects how precisely their trials determine sensitivity, accounting for trial count, performance level on the probit curve, and uncertainty in the criterion.

This variance is calculated using the large-sample Fisher information formula from Gourevitch and Galanter (1967). Unlike `sdt_moments()`, which evaluates that formula at raw empirical proportions (`var_gg`), `usdt_reliability()` evaluates it at the response probabilities predicted by the fitted hierarchical model.

Overall task reliability averages v_{ij} across participants, representing the expected proportion of true variance for a participant drawn at random from the sample.

When object includes bootstrap replicates from `usdt_boot()`, confidence intervals for reliability are computed automatically across all retained samples.

Value

An object of class `usdt_reliability` containing:

- `$tasks`: Overall reliability and variance components for each task.
- `$subjects`: Participant-level d' , error variances, and individual reliabilities.
- Bootstrap intervals for both components when available in object.

References

Gourevitch, V., & Galanter, E. (1967). A significance test for one parameter isosensitivity functions. *Psychometrika*, 32(1), 25–33. doi:[10.1007/BF02289402](https://doi.org/10.1007/BF02289402)

See Also

[hsdt\(\)](#), [usdt_boot\(\)](#), [sdt_moments\(\)](#)

Examples

```
# Contextual cuing data from Vadillo et al. (2025)
d <- usdt_data_tasks(
  direct    = vadillo_awareness,
  indirect  = vadillo_cuing,
  subject_col = "subj",
  condition_col = "condition",
  condition_levels = c(signal = "old", noise = "new"),
  response_col = list(direct = "judged.old", indirect = "rt"),
  response_levels = list(direct = c(signal = 1, noise = 0),
                        indirect = c(signal = "faster", noise = "slower")),
  dichotomize = list(direct = FALSE, indirect = TRUE)
```

```

)

r <- usdt_reliability(hsdt(d))
r

# Participant-level estimates (one row per subject and task)
head(r$subjects)

```

vadillo_awareness	<i>Awareness data from a probabilistic cuing experiment</i>
-------------------	---

Description

Trial-level direct-awareness data from Experiment 2 of Vadillo et al. (2025). The object reproduces the source CSV file without filtering or recoding and can be used as the direct input to `usdt_data_tasks()`.

Usage

```
vadillo_awareness
```

Format

A data frame with 6,656 rows, 104 participants, and 10 variables:

subj Participant identifier.
 file Original participant file name.
 trial Trial number.
 pattId Pattern identifier.
 judgm Awareness judgement on the original response scale.
 judged.old Whether the pattern was judged old (1) or new (0).
 condition Whether the pattern was old or new.
 offset Spatial-offset condition.
 color Colour condition.
 set.size Set-size condition.

Source

Vadillo, M. A., Malejka, S., & Shanks, D. R. (2025). Mapping the reliability multiverse of contextual cuing. *Journal of Experimental Psychology: Learning, Memory, and Cognition*. doi:10.1037/xlm0001410. Data retrieved from <https://osf.io/jp3gx/>.

Examples

```

data(vadillo_awareness)
str(vadillo_awareness)

```

vadillo_cuing*Cuing data from a probabilistic cuing experiment*

Description

Trial-level indirect-task data from Experiment 2 of Vadillo et al. (2025). The object reproduces the source CSV file without filtering or recoding and can be used as the indirect input to `usdt_data_tasks()`.

Usage

```
vadillo_cuing
```

Format

A data frame with 39,936 rows, 104 participants, and 13 variables:

`experiment` Experiment label.
`subj` Participant identifier.
`file` Original participant file name.
`trial` Trial number.
`block` Block number.
`epoch` Epoch number.
`pattId` Pattern identifier.
`acc` Response accuracy (1 = correct, 0 = incorrect).
`rt` Response time in milliseconds.
`condition` Whether the pattern was old or new.
`offset` Spatial-offset condition.
`color` Colour condition.
`set.size` Set-size condition.

Source

Vadillo, M. A., Malejka, S., & Shanks, D. R. (2025). Mapping the reliability multiverse of contextual cuing. *Journal of Experimental Psychology: Learning, Memory, and Cognition*. doi:10.1037/xlm0001410. Data retrieved from <https://osf.io/jp3gx/>.

Examples

```
data(vadillo_cuing)  
str(vadillo_cuing)
```

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